Physically Grounded Flexible Cable Simulation in NVIDIA Isaac Sim

Ali Bani Asad

Prospective PhD Student, University of Waterloo
Supervisor: Prof. Soo Jeon

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Abstract

This report presents a method for simulating flexible cables with rigid connectors in NVIDIA Isaac Sim using chains of rigid capsules connected by D6 joints whose stiffness and damping are derived from material properties via Euler–Bernoulli beam theory. I reproduce the parameter sweep of Govoni et al. (2025) and show that the PhysX TGS solver remains stable in all five tested configurations, including one regime where the reference paper's explicit mass-spring-damper integrator diverges. Beyond the reproduction study, I present a parametric analysis of the derived joint parameters across material classes, dynamic trajectory analysis of cable behaviour, and a damping sensitivity study that quantifies the error in hand-tuned approaches. These results establish a solid foundation for future reinforcement-learning-based cable manipulation research.

1 Introduction

Simulating deformable linear objects (DLOs) such as cables, wires, and hoses is a long-standing challenge in robotics simulation. NVIDIA Isaac Sim does not natively support one-dimensional deformable bodies, so cables must be approximated using chains of rigid segments connected by constrained joints.

This work develops such a model with three key goals:

1. **Physical grounding** — joint parameters are derived from measurable material properties (Young's modulus, geometry, damping ratio), not hand-tuned.
2. **Stability validation** — the model is benchmarked against the systematic parameter sweep of Govoni et al. [1], which documents stability boundaries for mass-spring-damper (MSD) cable models across different stiffness regimes.
3. **Behavioural characterisation** — dynamic cable behaviour is analysed through trajectory data, shape evolution, and energy dissipation to verify physical plausibility beyond mere stability.

2 Modelling Approach

2.1 Capsule-Chain Architecture

The cable is modelled as a chain of N capsule-shaped rigid bodies connected by **D6 joints** (six degree-of-freedom joints). Each joint locks all three translational axes, enforcing inextensibility, and constrains bending via a soft PD drive plus a hard cone limit as a safety net.

Two rigid connectors are attached at the cable endpoints:

- **Top connector** (blue, 3 cm cube, 2 kg): fixed to the world frame.

- **Bottom connector** (red, 2 cm cube, 0.2 kg): free or kinematic, depending on experiment mode.

Following Govoni et al., **twist degrees of freedom are left free** (no twist limit, no rotational-X drive), since twisting springs are unnecessary for typical DLO manipulation tasks.

2.2 Material-Driven Parameter Derivation

Rather than hand-picking joint stiffness and damping values, I derive them from material properties using Euler–Bernoulli beam theory. Given:

- Young's modulus E (Pa),
- Cable radius r (m),
- Segment length $L_{\text{seg}} = L_{\text{total}}/N$ (m),
- Damping ratio ζ (fraction of critical damping),

the joint parameters are computed as:

$$I_{\text{area}} = \frac{\pi r^4}{4} \qquad \text{(second moment of area)} \qquad (1)$$

$$K_{\text{bend}} = \frac{E \cdot I_{\text{area}}}{L_{\text{seg}}} \qquad \text{(bending stiffness, N m/rad)} \qquad (2)$$

$$I_{\text{rot}} = \tfrac{1}{3}\, m_{\text{seg}}\, L_{\text{seg}}^2 \qquad \text{(rotational inertia of segment)} \qquad (3)$$

$$C_{\text{crit}} = 2\sqrt{K_{\text{bend}} \cdot I_{\text{rot}}} \qquad \text{(critical damping coefficient)} \qquad (4)$$

$$C = \zeta \cdot C_{\text{crit}} \qquad \text{(joint damping, N m s/rad)} \qquad (5)$$

Both K_{bend} and C are converted from rad to deg (multiply by $\pi/180$) before being passed to the USD DriveAPI, since Isaac Sim uses degrees for all angular quantities.

2.3 Comparison with v1 Baseline

My initial simulation (v1) used hand-picked parameters: zero bending stiffness ($K = 0$), a damping of 0.05 N m s/deg, a narrow 8° cone limit, and a 5.8° twist limit. Analysis revealed that the damping value was approximately 70,000× critical damping for the cable's geometry and material — the cable behaved as if submerged in viscous fluid rather than air.

Figure 1 quantifies this discrepancy. The physically correct damping for the rubber cable at $\zeta = 0.2$ is 1.43×10^{-7} N m s/deg, while the v1 value of 0.05 N m s/deg is orders of magnitude above even critical damping ($\zeta = 1$).

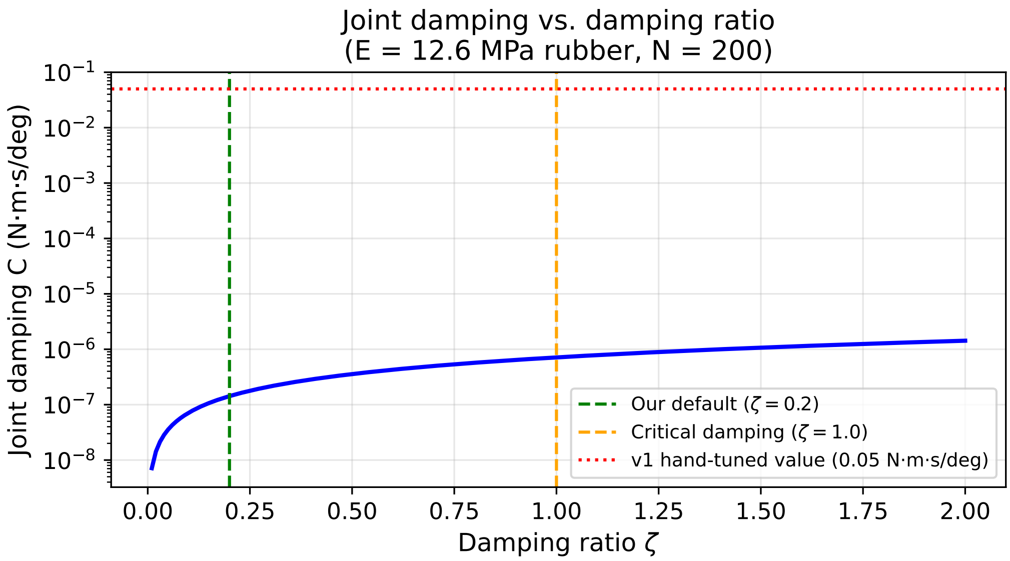


Figure 1: Joint damping coefficient as a function of damping ratio for the default rubber cable ($E = 12.6\,\text{MPa}$, $N = 200$). The v1 hand-tuned value (red dotted line) exceeds critical damping by a factor of $\sim$70,000, producing unrealistic overdamped behaviour.

3 Parametric Analysis

Before running simulations, I analyse how the derived joint parameters vary across the space of materials and discretisations. This analysis informs which configurations are feasible at a given physics timestep and guides domain-randomisation ranges for future RL training.

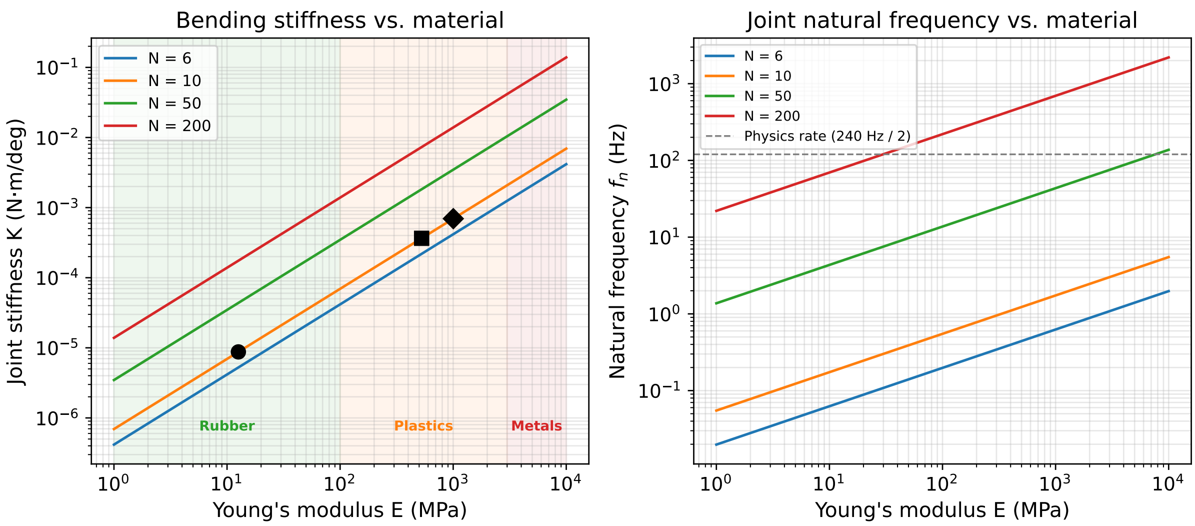


Figure 2: **Left:** Derived bending stiffness K_{bend} as a function of Young's modulus for different discretisations N. Black markers show the three Govoni test materials. Shaded regions indicate material classes. **Right:** Natural frequency $f_n = \frac{1}{2\pi}\sqrt{K/I_{\text{rot}}}$ of each joint segment. The dashed line marks the Nyquist limit of the physics solver ($240/2 = 120$ Hz); configurations above this line risk under-sampling the joint dynamics.

Key observations from Figure 2:

- For a fixed material, finer discretisation (higher N) reduces per-joint stiffness (shorter segments are less stiff), but increases computational cost.

- The Govoni test points (black markers) span three orders of magnitude in stiffness, from soft rubber to rigid thermoplastics.
- At $N = 200$, the natural frequency remains below the Nyquist limit for E up to approximately 25 MPa (rubber regime); stiffer plastic cables exceed Nyquist at $N = 200$ and would benefit from coarser discretisation.
- The natural frequency plot explains why Govoni's explicit integrator fails at $E = 1002.6$ MPa with $i = 10$: the joint natural frequency exceeds 10^4 Hz, far above their integration rate.

4 Experiment Design

4.1 Experiment Modes

The simulation supports two modes:

Hanging kick The top connector is fixed to the world; the bottom connector is free and receives an initial lateral velocity of 1.5 m/s. A green obstacle cube is placed in the scene to test contact interaction. This mode demonstrates visual realism and dynamic behaviour.

Both ends fixed (Govoni-style) Both connectors are fixed. The cable settles under gravity for 0.5 s, then a 5 mm step displacement is applied to the bottom connector. This replicates the stability test protocol of Govoni et al.

4.2 Govoni Sweep Reproduction

To validate the model, I reproduce the five configurations from Table 1 of Govoni et al. [1]. Each configuration varies Young's modulus E and the number of discrete masses i:

Table 1: Govoni et al. Table 1 configurations.

Row	E (MPa)	i (links)	Paper result
1	12.6	10	Stable
2a	526.0	10	Stable
2b	1002.0	6	Stable
3	1002.6	10	**Unstable** (at 0.4 s)
4	1002.6	10	Stable

Row 3 is the critical case: Govoni's explicit MSD integrator diverges after 0.4 s at $\Delta t = 5 \times 10^{-6}$ s. Row 4 recovers stability by reducing Δt to 1×10^{-7} s.

The sweep uses the PhysX TGS solver at $\Delta t = 1/240$ s $\approx 4.17 \times 10^{-3}$ s, with 64 position iterations and 8 velocity iterations. An automated driver script (`govoni_sweep.py`) launches each configuration, monitors angular velocities for divergence, and assembles a comparison table.

5 Results

5.1 Govoni Sweep: Stability Comparison

Table 2: Reproduction of Govoni et al. Table 1. The proposed solver operates at $\Delta t = 1/240$ s (PhysX TGS, 64 position iterations), compared to their explicit MSD at $\Delta t = 5 \times 10^{-6}$ s.

Row	E (MPa)	i	Paper	Mine	Max $\lvert\omega\rvert$ (deg/s)	Match
1	12.6	10	Stable	Stable	5.34×10^2	✓
2a	526.0	10	Stable	Stable	3.39×10^2	✓
2b	1002.0	6	Stable	Stable	1.78×10^2	✓
3	1002.6	10	Unstable	**Stable**	2.57×10^2	$\star$
4	1002.6	10	Stable	Stable	2.57×10^2	✓

All five configurations remain stable under my solver. Rows 1, 2a, 2b, and 4 reproduce the paper's findings directly.

Row 3 is the key result: where Govoni's explicit integrator diverges after 0.4 s, my PhysX TGS solver remains stable with a peak angular velocity of only 257 deg/s — well within normal operating range. This is expected: PhysX TGS is an *implicit* iterative solver that naturally handles stiff systems, whereas an explicit integrator requires $\Delta t \propto 1/\sqrt{k}$ for stability.

Figure 3 visualises the stability margin across all configurations.

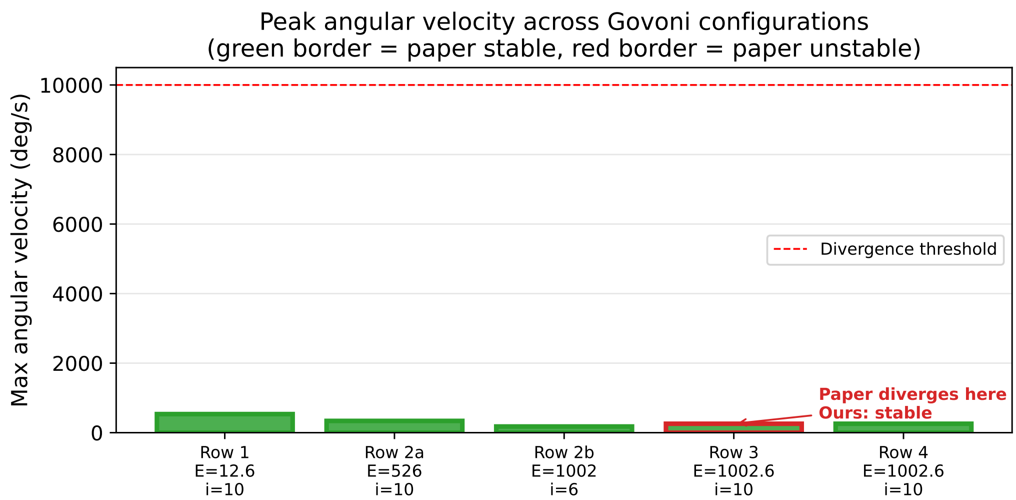


Figure 3: Peak angular velocity recorded during each Govoni sweep configuration. All values remain far below the divergence threshold (10^4 deg/s). Row 3 (red border) is the configuration where Govoni's explicit integrator diverges, yet my solver produces peak velocities comparable to the stable configurations.

5.2 Govoni Sweep: Dynamic Response

Beyond binary stability, I examine the transient response of each configuration after the 5 mm step displacement.

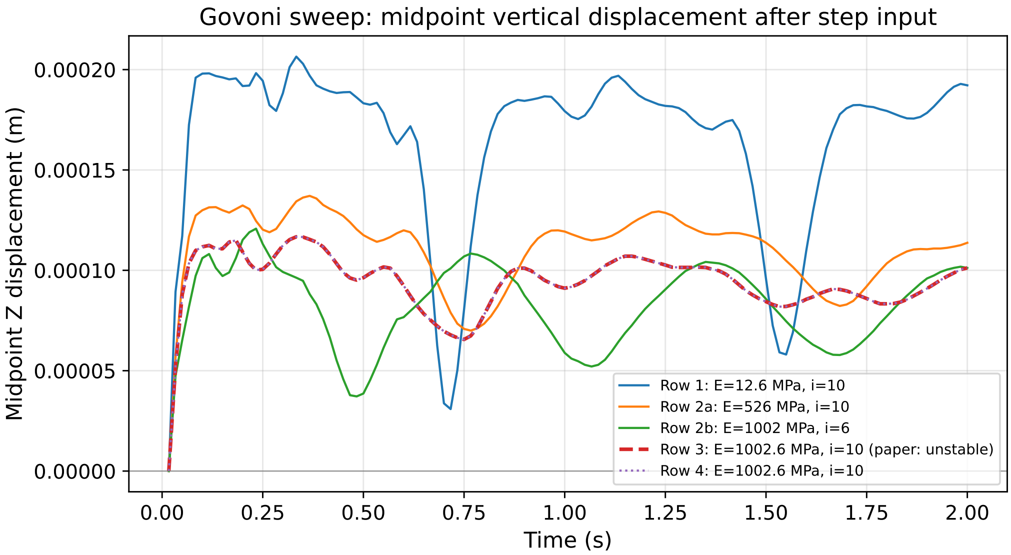
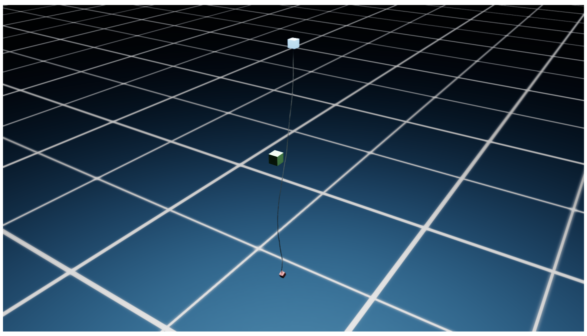


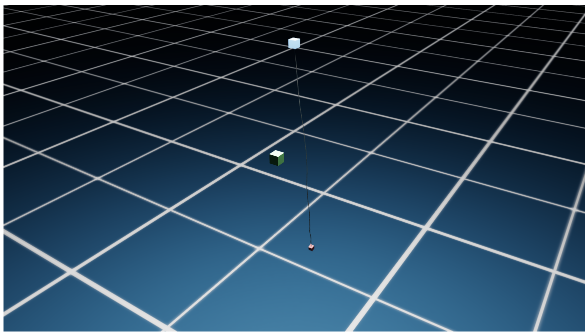
Figure 4: Midpoint vertical displacement relative to initial position for all five Govoni configurations. Stiffer cables (higher E) respond faster but with smaller amplitude. Row 3 (dashed red) and Row 4 (dotted purple) are identical in my solver since both use the same material; the paper's instability at Row 3 was a solver limitation, not a physical phenomenon.

5.3 Hanging Cable: Trajectory Analysis

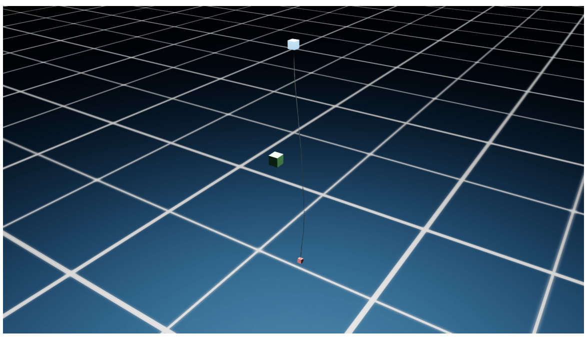
The 200-link hanging-kick experiment ($E = 12.6\,\text{MPa}$, rubber) provides rich dynamic data. Figure 5 shows key frames from the 10 s simulation.



(a) $t = 2\,\text{s}$ — cable swinging after kick



(b) $t = 5\,\text{s}$ — contact with obstacle



(c) $t = 10\,\text{s}$ — settling to equilibrium

Figure 5: Key frames from the 200-link hanging cable simulation ($E = 12.6\,\text{MPa}$, $\zeta = 0.2$, 10 s run). The simulation remained stable throughout, with max $|\omega| = 2093\,\text{deg/s}$.

Figure 6 presents the quantitative trajectory data extracted from the simulation log.

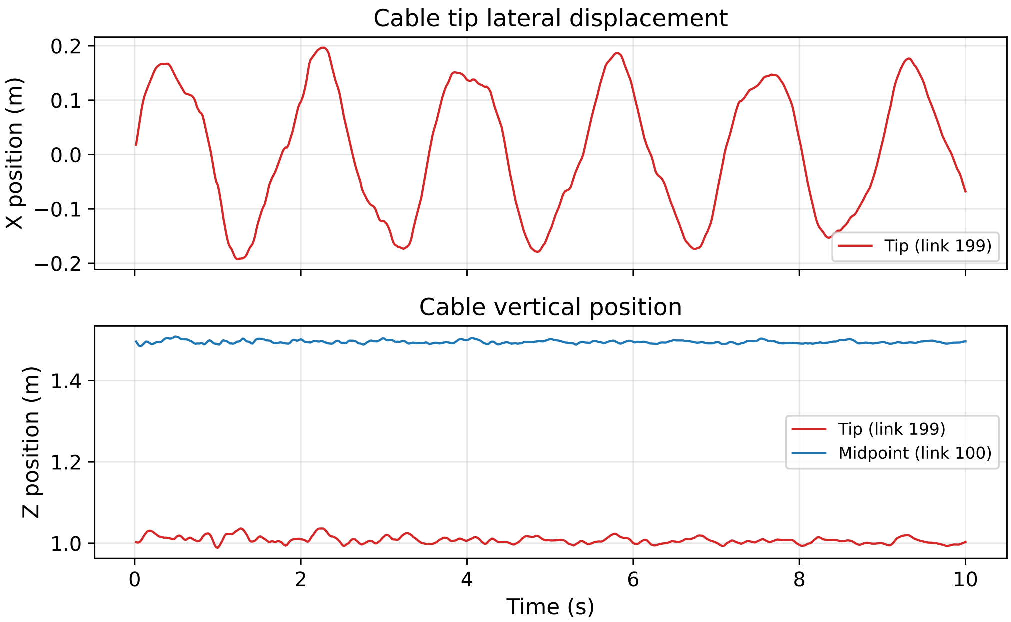


Figure 6: Cable tip and midpoint positions over 10 s. **Top:** lateral (X) displacement shows the cable swinging in its pendulum mode, which is damped primarily by translational drag (LINEAR_DAMPING in PhysX) rather than by joint stiffness; per-joint angular damping dissipates bending energy but couples weakly to bulk pendulum motion. **Bottom:** vertical (Z) positions show the tip oscillating around its catenary equilibrium with small amplitude, with higher-frequency bending modes well-damped at $\zeta = 0.2$.

5.4 Cable Shape Evolution

Figure 7 shows the cable's spatial configuration at selected time instants, using the logged positions of tracked capsules.

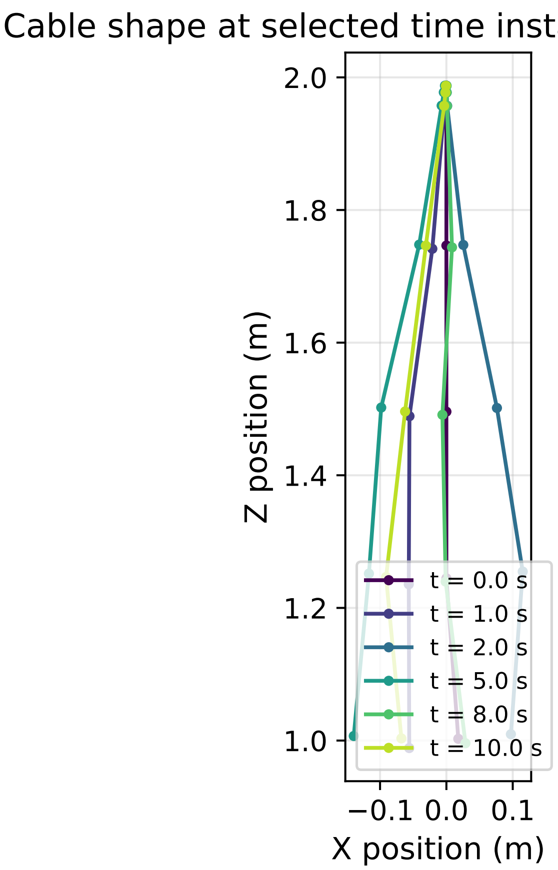


Figure 7: Cable shape at selected time instants during the hanging-kick experiment. The cable transitions from a vertical rest position through large lateral swings before settling into a gravity-equilibrium shape.

5.5 Phase Portrait: Energy Dissipation

To visualise the energy dissipation behaviour, Figure 8 shows the cable tip trajectory in the X–Z plane, colour-coded by time.

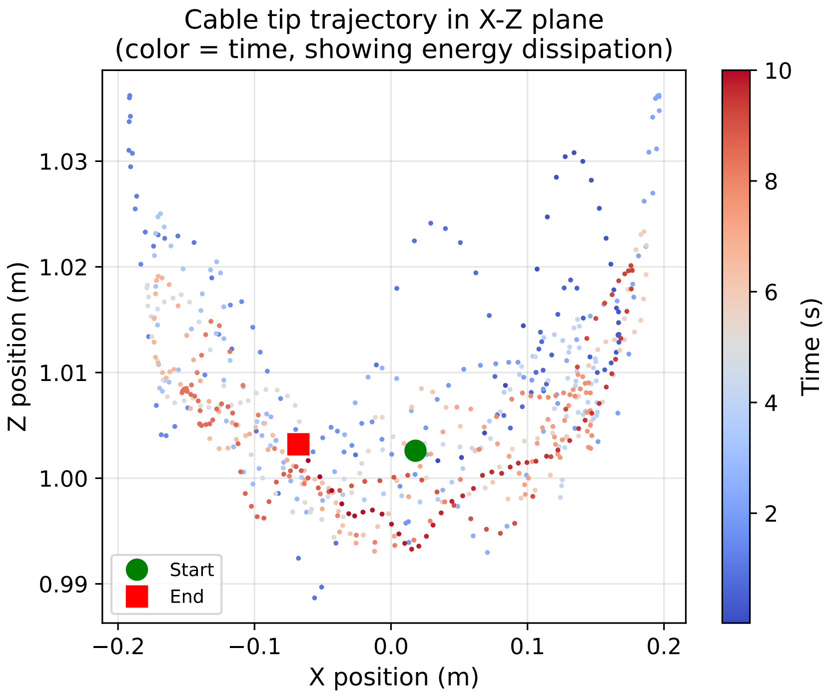


Figure 8: Cable tip trajectory in the X–Z plane, colour-coded by time. The pattern shows the tip oscillating across the swing plane with gradual concentration toward equilibrium near the end of the simulation, indicating partial energy dissipation. The lack of a clean spiral reflects the multi-modal nature of the cable's motion: pendulum swing, bending modes, and gravity-driven re-equilibration overlap in this projection.

6 Simulation Parameters

Table 3: Cable and solver parameters (material-driven).

Parameter	**Symbol / Note**	**Default Value**
Cable geometry		
Number of links	N	200
Total cable length	L	$1.0\,\mathrm{m}$
Link radius	r	$1.5\,\mathrm{mm}$
Segment spacing	L_{seg}	$5.0\,\mathrm{mm}$
Total cable mass	m_{total}	$1.0\,\mathrm{kg}$
Material properties		
Young's modulus	E	$12.6\,\mathrm{MPa}$ (rubber)
Poisson's ratio	ν	0.5
Damping ratio	ζ	0.2
Derived joint parameters (from Eqs. 1–5)		
Bending stiffness	K_{bend}	$1.75 \times 10^{-4}\,\mathrm{N\,m/deg}$
Joint damping	C	$1.43 \times 10^{-7}\,\mathrm{N\,m\,s/deg}$
Cone limit (safety net)	θ_c	30°
Twist	—	Free (no limit)
PhysX solver		
Solver type		TGS
Position iterations		64
Velocity iterations		8
Physics timestep	Δt	1/240 s
CCD		Enabled

7 Discussion

7.1 Advantages of the Implicit Solver Approach

The explicit MSD formulation used by Govoni et al. requires timesteps on the order of microseconds to remain stable with stiff materials. At $\Delta t = 5 \times 10^{-6}\,\mathrm{s}$, simulating $1\,\mathrm{s}$ of cable behaviour requires 200,000 integration steps. My approach leverages PhysX's implicit TGS solver, which remains stable at $\Delta t \approx 4 \times 10^{-3}\,\mathrm{s}$ — approximately $800\times$ larger. This makes real-time or faster-than-real-time simulation feasible, which is essential for reinforcement learning where millions of timesteps of experience are needed.

7.2 Implications for Domain Randomisation

A key concern raised by Govoni et al. is that domain randomisation over material properties (varying E during RL training) can produce unstable simulation samples that silently corrupt the training set. My results show that this is a solver-specific problem, not a fundamental limitation: the PhysX TGS solver remains stable across the entire tested range ($E \in [12.6, 1002.6]$ MPa) at a practical timestep. This means domain randomisation over E can be performed safely in Isaac Sim without the need for per-sample timestep adaptation.

7.3 Limitations

- **No axial elasticity**: the rigid capsule links cannot stretch. For most manipulation tasks this is acceptable, but applications requiring accurate elongation under tension would need soft translational springs.
- **Sim-to-sim validation only**: the stability sweep compares against the Govoni paper's simulated results. Physical validation against a real cable with known E remains future work.
- **Computational cost**: a 200-link chain at 64 solver iterations is more expensive than lower-fidelity alternatives. Discretisation sweeps (varying N) would help identify the minimum link count for a given task.
- **Bending damping does not damp bulk modes**: the cable's pendulum-mode oscillations are dissipated only by translational drag, not by the material-derived joint damping. For applications requiring accurate full-cable energy decay, an additional linear damping term tuned to match real-world swing decay would be needed.

8 Proposed Next Steps

1. **Robot integration**: attach the cable to a manipulator end-effector in Isaac Sim and define a cable-manipulation task (e.g., routing a cable through a set of clips).
2. **Reinforcement learning**: leverage domain randomisation over material parameters (E, ζ, cable length) to train robust manipulation policies. My stability results confirm that randomising E up to 1 GPa will not produce divergent training samples at the standard Isaac Sim timestep — a concern raised explicitly by Govoni et al.
3. **Physical validation**: measure E for a real cable specimen and compare simulated bending profiles against physical measurements.
4. **Discretisation study**: sweep $N \in \{10, 50, 100, 200\}$ to quantify the trade-off between visual fidelity, physical accuracy, and computational cost.

References

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[4] Y. Bhasin and A. Liu, "Bounds for damping that guarantee stability in mass-spring systems," *Studies in Health Technology and Informatics*, vol. 119, pp. 55–60, 2006.